

Read all questions carefully and answer them completely.

1. Our bodies produce cholesterol as a building block for cell walls, hormones, and bile salts. There are two different types of cholesterol, low-density lipoprotein (LDL) and high-density lipoprotein (HDL), and high levels of LDL cholesterol are associated with risk of heart attack and stroke.

Janice is interested in comparing the amount of reduction in LDL levels in patients with elevated levels using either diet and exercise (a "lifestyle" change), or using statin drugs. She recruits 30 volunteers with elevated LDL counts and randomly places them into one of three groups: controls (who are given neither diet or drug interventions), lifestyle (who are given a low-LDL diet and are tasked with exercising for 30 minutes 3 times per week) and statin (who are given prescriptions for statin drugs). After one month each volunteer is tested again for LDL levels, and the difference in LDL count is used to measure the reduction in LDL for each person.

The data for the experiment are in question1.MTW. "Group" gives the treatment group, and "Change in LDL" gives the amount of change in LDL count for each person (negative numbers mean a reduction in LDL during the study).

A. What procedure will you use to compare the mean LDL levels among these three groups?

ANOVA

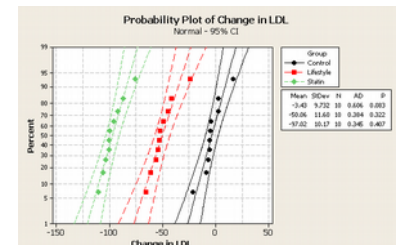
B. Test for HOV, and report the test statistic and p-value for Levene's test in the table to the right.

HOV

Levene's test stat.	0.13
p	0.876
HOV? (yes/no)	Yes

C. Do you have normality? How did you assess this (you can use either AD tests or residual plots, state which you used)?

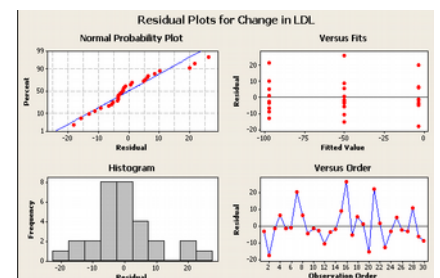
If you used AD tests, the graphs are to the right. The residual plots show no reason to doubt normality.



D. What is the null hypothesis for this test?

$$H_0: \mu_{\text{control}} = \mu_{\text{statin}} = \mu_{\text{lifestyle}}$$

E. Report the ANOVA table here:



Source	DF	SS	MS	F	p
Group	2	43796	21898	197.49	< 0.001
Error	27	2994	111		
Total	29	46789		Reject or retain?	Reject

F. Can you tell from the ANOVA table whether you should reject or retain the null hypothesis? Can you tell from the ANOVA table alone which treatment groups differ from one another? Why or why not?

We reject the null because $p < 0.05$ but we can't tell which groups are different until we do a Tukey post-hoc procedure.

G. Which treatment groups differ from one another? Which treatment was most effective at reducing LDL count?

All three are different from one another. The statin drugs showed the biggest amount of decrease in LDL.

2. It is thought that in addition to the amount of LDL in the blood, the ratio of HDL (high-density lipoprotein cholesterol) to LDL cholesterol is also important in heart health. Having a large amount of LDL compared with the amount of HDL is bad, even if the LDL level is fairly low.

Janice is concerned that statins may have a bigger effect on HDL than on LDL cholesterol, such that even though statins may reduce LDL levels they may leave patients with unhealthy HDL/LDL ratios. To test this, Janice wished to compare the HDL/LDL ratio of the two treatment groups from the experiment above. She takes the HDL level for each patient and divides it by the LDL level, to obtain an HDL/LDL ratio. Bigger numbers are better. If the ratios are no different on average between the groups, or if the statin group is higher than the lifestyle group, then statins do not produce unhealthy ratios compared to diet and exercise.

In the previous experiment, the HDL/LDL ratios for each patient in the lifestyle and statin groups are given in file question2.MTW.

A. Are the data normally distributed? Report the AD and p-values for the tests.

Lifestyle is, statin is not.

B. Apply the appropriate transformation and re-test for normality.

C. Test for HOV on the data in which both groups meet the normality assumption and report it in the HOV table to the right.

D. Test for differences in mean HDL/LDL ratio using the transformed data.

$t = \underline{1.92}$ $df = \underline{18}$ $p = \underline{0.071}$ Reject H_0 (yes/**no**)?

E. Interpret the results of the hypothesis test - do statin drugs significantly change the HDL/LDL ratio relative to lifestyle changes?

There is no difference in ratio between the two.

Normality - untransformed

	Lifestyle	Statin.
AD	0.508	0.731
p	0.150	0.038
Normal? (yes/no)	Yes	No

Normality - transformed

	Lifestyle	Statin.
AD	0.133	0.317
p	0.968	0.477
Normal? (yes/no)	Yes	Yes

HOV

Levene's test stat.	1.28
p	0.273
HOV? (yes/no)	Yes

F. Back-transform the following values from the log scale to the original scale of the data, then interpret what each one means.

Statistic	On log scale	Back-transformed	Interpretation of back-transformed value
Lifestyle mean	1.047	2.85	<i>Lifestyle geometric mean</i>
Statin mean	0.433	1.54	<i>Statin geometric mean</i>
Difference	0.614	1.85	<i>Ratio of lifestyle GM to statin GM</i>

3. File question3.MTW has the same HDL/LDL ratio as in Question 2, in an unstacked arrangement. Use these data to run a Mann-Whitney U test.

A. $W = 127.0$ $p = 0.1041$ Reject H_0 (yes/**no**)?

B. Report the medians for lifestyle and statin drugs here.

Lifestyle median = 2.701 Statin median = 1.441

C. What is the interpretation of this result? Are the medians significantly different?

The medians are not significantly different from one another, so the statins are not changing the median HDL/LDL ratios.

4. Finally, Janice decides to run a randomization test on the HDL/LDL ratios for comparison with the other two tests she already ran. She randomly shuffles her data and calculates differences between the arithmetic means for the random groups 9,999 times.

The differences between the randomly shuffled groups are in file question4.MTW. The differences are in the first column, and the second column indicates whether the differences exceed the observed difference between the means of 1.295.

A. What is the null hypothesis for this test? Be sure to identify what kind of measure of central tendency is being tested.

$H_0: \mu_{control} = \mu_{statin}$, and the measure of central tendency is the arithmetic mean.

B. What is the p-value for this test? Show your work for possible partial credit.

$$\frac{\text{exceeds} + 1}{\text{total} + 1} = \frac{1747 + 1}{9999 + 1} = 0.1748$$

C. Interpret the result - is there a difference in HDL/LDL ratio between the groups for this test?

There is no difference in mean HDL/LDL ratio. Statins don't change the ratio.

5. Compare the three analyses of differences in HDL/LDL ratio. What was the measure of central tendency used for each, and what was the null hypothesis for each? For the transformed data explain the test in terms of the back-transformed values - how did transformation change the measure of central tendency and the null hypothesis for the t-test?

The test of transformed HDL/LDL ratios compared the ratio of geometric means to a value of 1 (so, $H_0: GM_{lifestyle}/GM_{statin} = 1$). The Mann-Whitney U test compared the medians (so, $H_0: \eta_{lifestyle} = \eta_{statin}$). The randomization test compared differences between arithmetic means (so, $H_0: \mu_{statin} = \mu_{lifestyle}$). None of the comparisons were significant. The back-transformed values were geometric means for the groups, and the back-transformed difference was the lifestyle geometric mean divided by the statin geometric mean, which was $2.85/1.54 = 1.85$. This was not significantly different from 1.

Optional probability question. If your score is higher on this page than on the probability question on Midterm 1, this score will replace your score on that question.

The data to the right represents the results of an assessment of a screening test for streptococcal bacterial infection. The first column is the test result (positive means the bacteria are present, negative means they are absent), and the actual strep infection status is given in the second and third column.

Test result	Strep infection status		Total
	Strep	No strep	
Positive	64	36	100
Negative	321	9679	10000
Total	385	9715	10100

A. What is the probability of a positive test result (that is, $P(\text{positive})$)? Is this a conditional or marginal probability?

$100/10100 = 0.0099$, conditional probability.

B. What is the probability of having strep (that is $P(\text{Strep})$)?

$385/10100 = 0.038$

C. What is the probability of having strep and getting a positive test result ($P(\text{Strep and Positive})$)?

$64/10100 = 0.006$

D. What is the probability of getting a positive result given that you have strep ($P(\text{positive} | \text{strep})$)? Is this a conditional or marginal probability?

$64/385 = 0.166$, conditional probability.

E. What is the probability of having strep given that you get a positive result ($P(\text{strep} | \text{positive})$)? Would you need to reduce false positives or false negatives to make this probability higher?

$64/100 = 0.64$, and you would need to reduce false positives to make this increase.

F. Is strep infection status independent of the test result? Which two probabilities did you compare to determine this?

You need to compare a marginal probability to a conditional probability of the same thing. We have two we could use, $P(\text{strep})$ compared to $P(\text{strep} | \text{positive})$, or $P(\text{positive})$ compared to $p(\text{positive} | \text{strep})$. In both cases the marginal is different from the conditional, so they are not independent.